



Life Cycle Assessment 2

ENGEN370 – Mark Lay



LCA Quarry

Quarry production: 3200 tons gravel per day

Functional unit: ton of gravel

Equipment used	Quantity	Hours used per day	Fuel consumption per unit (L/hr)	Electricity consumption (kWhr/ton)
Hydraulic drilling rig on excavator	1	2	10	
20 ton excavator with 3 ton bucket	2	10	20	
Cat 936E loader with 9 ton bucket	2	10	13.2	
40 ton hauler	8	10	27	
Crusher		10		2.2
Classification		10		1
Belts		10		0.1

Fuel type	Density (kg/L)	CV (MJ/kg)	Formula	MW (g/mol)	Mols CO2 produced /Mol Fuel	Combustion (kg CO2 /kg fuel)	Production (kg CO2 /kg fuel)	Total (kg CO2 /kg fuel)
Diesel	0.832	44.8	C12H24	168.96	12	3.13	0.096	3.22
Biodiesel	0.875	37.27	C19H34O2	295.36	19	2.83	-1.53	1.30
Methane		52	CH4	16	1	2.75	0.01	2.76
Coal		33	C8.8H0.5O0.3	116.92	8.8	3.31	0.01	3.32

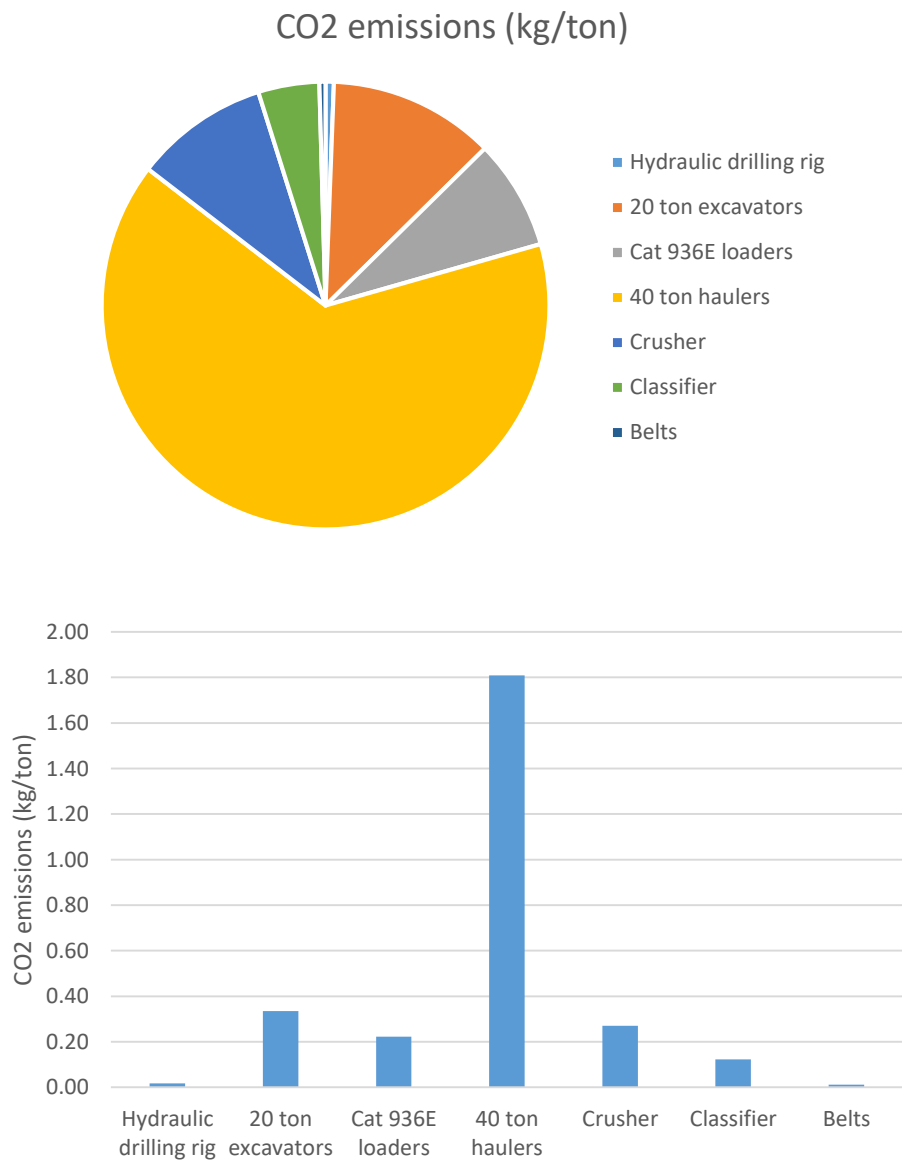
LCA Quarry

Huntly Power Station	Power (MW)	Efficiency	Capacity (%)	Actual (MW)	Thermal (MW)	Fuel type	Fuel consumption (kg/s)	CO2 emissions (kg/s)	CO2 emissions (kg/MJ)
Overall	900		65	585					
CC	400	0.56	100	400	714	Methane	13.7	37.9	0.095
Rankine	500	0.3	37	185	617	Coal	18.7	62.1	0.336
Total				585				100.0	0.171

Fraction renewable energy NZ = 0.8

LCA Quarry Mass Allocation

	NRE (MJ/day)	NRE (MJ/ton)	CO2 emissions (kg/day)	CO2 emissions (kg/ton)
Hydraulic drilling rig			53.6	0.02
20 ton excavators			1071.9	0.33
Cat 936E loaders			710.1	0.22
40 ton haulers			5788.5	1.81
Crusher	5068.8	1.584	866.3	0.27
Classifier	2304	0.72	393.8	0.12
Belts	230.4	0.072	39.4	0.01
Total	7603.2	2.376	8923.7	2.79



LCA Economic Allocation

NRE (MJ/\$)	CO2 (kg/\$)
0.101	0.119

Product	Price (\$/tonne)	Fraction	Mass (tonnes)	Revenue (\$)	NRE (MJ)	CO2 (kg)	NRE (MJ/tonne)	CO2 (kg/tonne)
Topcourse Domestic 20mm	15.8	0.015	48	758.4	76.65	89.96	1.60	1.87
Topcourse Standard 20mm	20.64	0.025	80	1651.2	166.87	195.85	2.09	2.45
Basecourse Domestic 40mm	13.37	0.025	80	1069.6	108.10	126.87	1.35	1.59
Basecourse Standard 40mm	17.69	0.025	80	1415.2	143.02	167.86	1.79	2.10
Road Metal 65mm	11.54	0.15	480	5539.2	559.80	657.02	1.17	1.37
Road Metal 65mm M3 (NZTA Specification)	19.85	0.3	960	19056	1925.83	2260.29	2.01	2.35
Bedding 7mm	19.24	0.05	160	3078.4	311.11	365.14	1.94	2.28
Bedding 10mm	22.99	0.05	160	3678.4	371.75	436.31	2.32	2.73
Bedding 14-5mm	27.36	0.05	160	4377.6	442.41	519.24	2.77	3.25
Bedding 20mm	27.36	0.05	160	4377.6	442.41	519.24	2.77	3.25
Filter B 100-250mm	32.07	0.08	256	8209.92	829.71	973.80	3.24	3.80
Cribwall Backfill 40-20mm	27.24	0.03	96	2615.04	264.28	310.18	2.75	3.23
Cribwall Backfill 80-40mm	21.92	0.03	96	2104.32	212.67	249.60	2.22	2.60
Sealing Chip Grade 2	46.23	0.02	64	2958.72	299.01	350.94	4.67	5.48
Sealing Chip Grade 3	46.23	0.02	64	2958.72	299.01	350.94	4.67	5.48
Sealing Chip Grade 4	46.23	0.02	64	2958.72	299.01	350.94	4.67	5.48
Sealing Chip Grade 5	46.23	0.02	64	2958.72	299.01	350.94	4.67	5.48
Sealing Chip Grade 6	46.23	0.02	64	2958.72	299.01	350.94	4.67	5.48
Concrete Aggregate 14-5mm	39.7	0.01	32	1270.4	128.39	150.69	4.01	4.71
Concrete Aggregate 20-10mm	38.7	0.01	32	1238.4	125.15	146.89	3.91	4.59
Total		1	3200	75233.28	7603.20	8923.66	2.38	2.79

LCA Sensitivity

Fuel consumption	L/hr	Lower (L/hr)	Upper (L/hr)	Base (CO2 kg/ton)	Lower (CO2 kg/ton)	Upper (CO2 kg/ton)	Lower (%)	Upper (%)
20 ton excavator	20	18	22	2.79	2.75	2.82	-1.43	1.08
Cat 936E loader with 9 ton bucket	13.2	11.9	14.6	2.79	2.77	2.81	-0.72	0.72
Hydraulic drilling rig	10	9	11	2.79	2.79	2.79	0.00	0.00
40 ton hauler	27	24.3	29.7	2.79	2.61	2.97	-6.45	6.45

Electrical	kWhr/tonne	Lower	Upper	Base (CO2 kg/ton)	Lower (CO2 kg/ton)	Upper (CO2 kg/ton)	Lower (%)	Upper (%)
Crusher	2.2	1.98	2.42	2.79	2.76	2.82	-1.08	1.08
Classifier	1	0.9	1.1	2.79	2.78	2.8	-0.36	0.36
Belts	0.1	0.09	0.11	2.79	2.79	2.79	0.00	0.00
	Fraction	Lower	Upper	Base (CO2 kg/ton)	Lower (CO2 kg/ton)	Upper (CO2 kg/ton)	Lower (%)	Upper (%)
Renewable electricity	0.8	0.72	0.88	2.79	2.95	2.63	5.73	-5.73

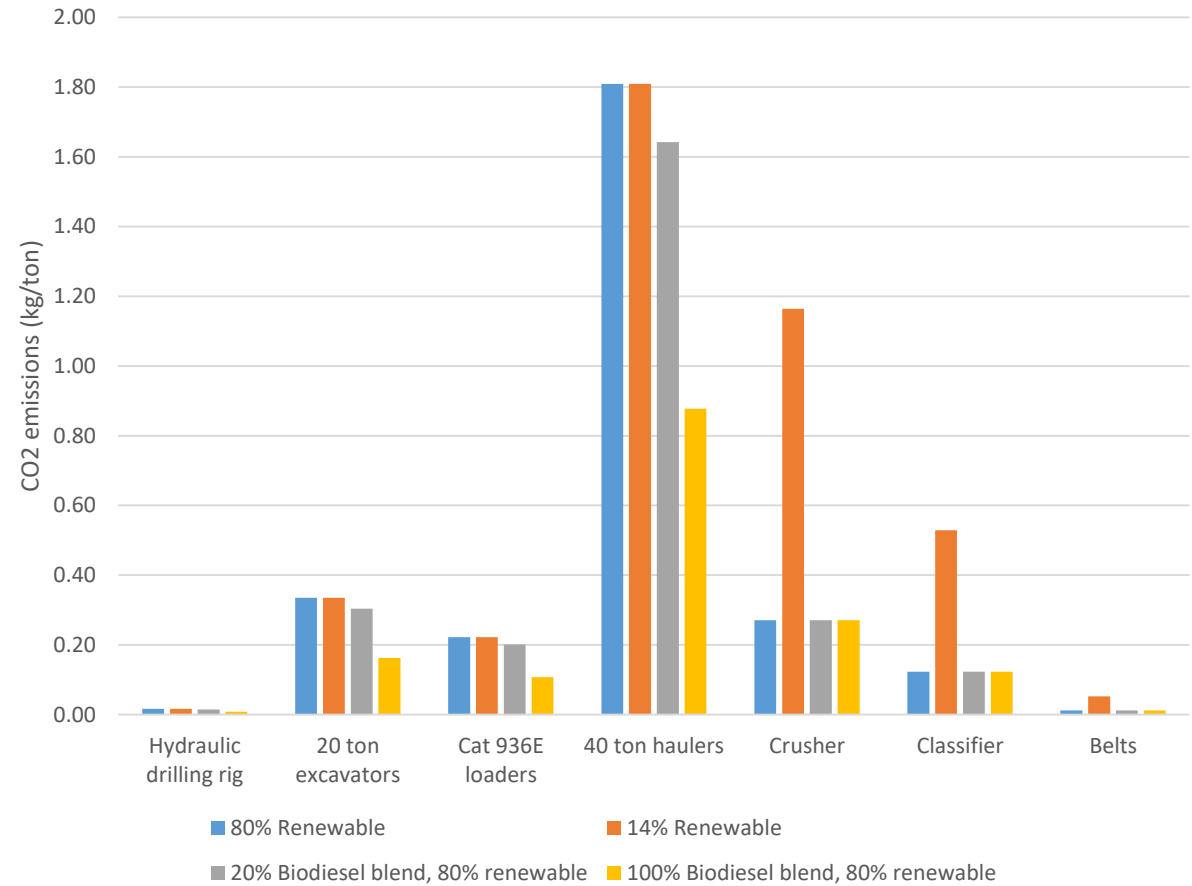
LCA Sensitivity

Fuel type	kg CO2/kg fuel production	Lower	Upper	Base (CO2 kg/ton)	Lower (CO2 kg/ton)	Upper (CO2 kg/ton)	Lower (%)	Upper (%)
Diesel	0.096	0.0864	0.1056	2.79	2.78	2.8	-0.36	0.36
Biodiesel	-1.53	-1.377	-1.683	2.79	2.79	2.79	0.00	0.00
Methane	0.01	0.009	0.011	2.79	2.79	2.79	0.00	0.00
Coal	0.01	0.009	0.011	2.79	2.79	2.79	0.00	0.00
Fuel type	CV (MJ/kg)	Lower	Upper	Base (CO2 kg/ton)	Lower (CO2 kg/ton)	Upper (CO2 kg/ton)	Lower (%)	Upper (%)
Methane	52	46.8	57.2	2.79	2.81	2.77	0.72	-0.72
Coal	33	29.7	36.3	2.79	2.82	2.77	1.08	-0.72
Huntly Power Station	Capacity	Lower	Upper	Base (CO2 kg/ton)	Lower (CO2 kg/ton)	Upper (CO2 kg/ton)	Lower (%)	Upper (%)
Capacity	65	58.5	71.5	2.79	2.75	2.82	-1.43	1.08

What if?

- 20% or 100% biodiesel blend
- 14% renewable electricity

Situation	CO2 emissions (kg/ton)
80% renewable	2.79
14% renewable	4.13
20% biodiesel blend, 80% renewable	2.57
100% biodiesel blend, 80% renewable	1.56



Monte Carlo simulation

- <https://www.youtube.com/watch?v=BQv2Uyea8i4>
- Monte Carlo Simulations in Excel without 3rd Party Add-ins
- 15 variables for the quarry that exist between a range
 - Uniformly distributed
 - Some kind of probability distribution
- Monte Carlo simulation
 - Randomly select those variables
 - Calculate a result
 - Repeat (e.g. up to 10,000 times or more)
 - Gives a probability of a result happening
 - Generate a histogram

Monte Carlo simulation

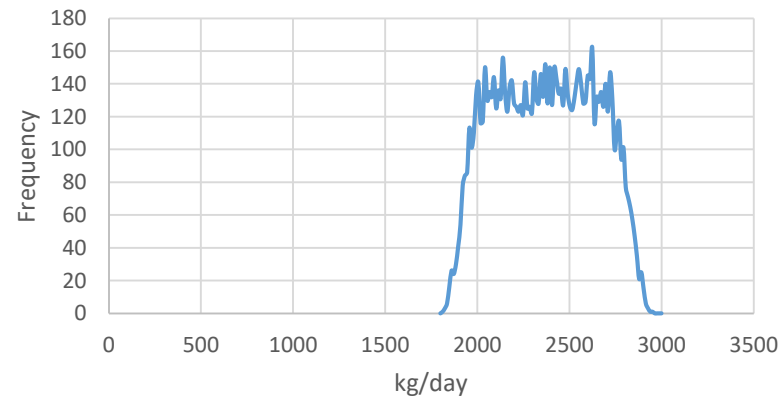
- In Excel use for each variable
 - Randbetween(lower value, upper value) for uniform distribution
 - Norminv(Rand(),mean value, standard deviation) for normal distribution
 - Likes whole positive numbers, not fractions
 - To display results of all calculations:
 - Set up a table with columns simulation# and the output
 - Rows 1:10000
 - Select the whole table
 - Use Data, What-if analysis, Data table
 - Row input cell – leave empty
 - Column input cell – select a cell that is empty and won't have any data
 - Excel will fill up the table with the results from each simulation

Monte Carlo simulation

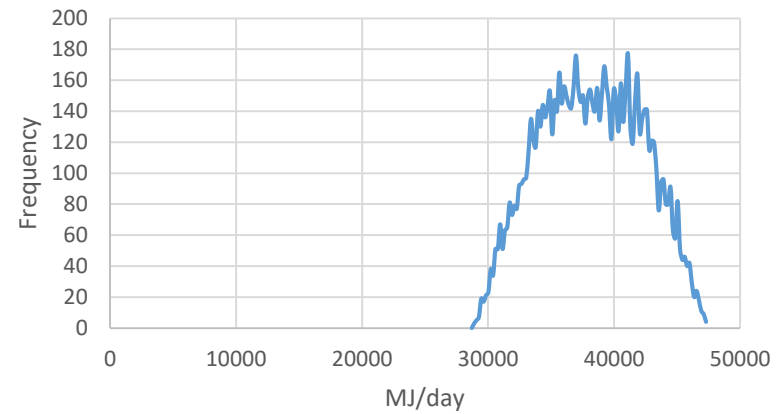
- To generate histograms
 - Turn on the Analysis ToolPak add-in
 - Create a table with the ranges you are collecting frequencies of values between
 - Use Data, Data Analysis, Histogram to generate the Histogram from the simulation results

Monte Carlo Simulation

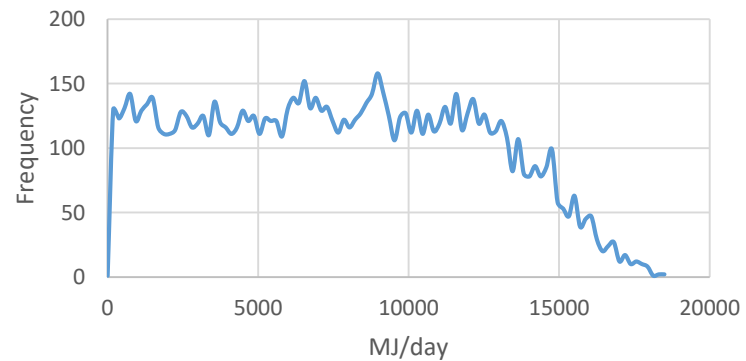
Fuel use



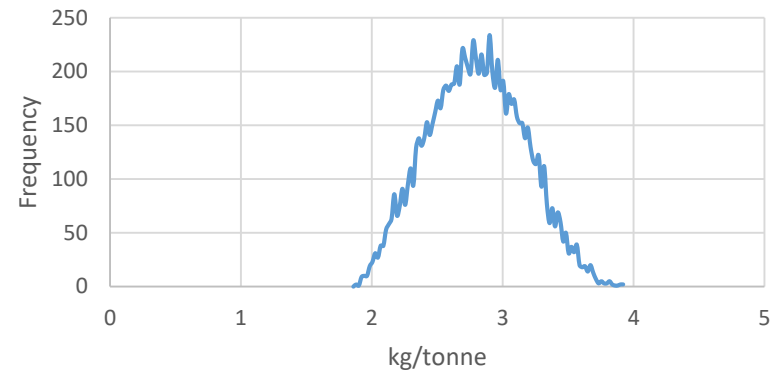
Electricity use



Non renewable electricity use



CO2



Monte Carlo Simulation

	Fuel use (kg)	Electricity (MJ)	Non renewable electricity (MJ)	CO2 emissions (kg/day)	NRE (MJ/ton)	CO2 (kg/ton)	NRE (MJ/\$)	CO2 (kg/\$)
Mean	2362.96	37998.23	7589.86	8896.77	2.37	2.78	0.10	0.12
Median	2367.51	38033.28	7427.04	8888.42	2.32	2.78	0.10	0.12
Min	1801.81	28748.16	0.00	5799.91	0.00	1.81	0.00	0.08
Max	2933.60	47287.30	18479.85	12394.94	5.77	3.87	0.25	0.16
Std. Dev.	264.82	4001.04	4477.03	1172.47	1.40	0.37	0.06	0.02

Data Quality

(Ref: Energy Efficiency and Conservation Authority, Life Cycle Assessment of Electric Vehicles
Final Report, 2015)

- Reliability (actual, estimated, calculated, opinion)
- Completeness (complete – all sites, some sites, one site, unknown)
- Temporal correlation (current, old, really old)
- Geographical correlation (actual, larger or smaller area, different country)
- Technology (data from that under study, related, different scale, lab)
- Sample size (bigger is better)

Pedigree matrix



Score:	1	2	3	4	5
U1 Reliability	Verified data based on measurements	Verified data partly based on assumptions OR non-verified data based on measurements	Non-verified data partly based on qualified estimates	Qualified estimate (e.g. by industrial expert); data derived from theoretical information (stoichiometry, enthalpy, etc.)	Non-qualified estimate
	1.00	1.05	1.10	1.20	1.50
U2 Completeness	Representative data from all sites relevant for the market considered over an adequate period to even out normal fluctuations	Representative data from >50% of the sites relevant for the market considered over an adequate period to even out normal fluctuations	Representative data from only some sites (<<50%) relevant for the market considered OR >50% of sites but from shorter periods	Representative data from only one site relevant for the market considered OR some sites but from shorter periods	Representativeness unknown or data from a small number of sites AND from shorter periods
	1.00	1.02	1.05	1.10	1.20
U3 Temporal correlation	Less than 3 years of difference to our reference year (2000)	Less than 6 years of difference to our reference year (2000)	Less than 10 years of difference to our reference year (2000)	Less than 15 years of difference to our reference year (2000)	Age of data unknown or more than 15 years of difference to our reference year (2000)
	1.00	1.03	1.10	1.20	1.50
U4 Geographical correlation	Data from area under study	Average data from larger area in which the area under study is included	Data from smaller area than area under study, or from similar area		Data from unknown OR distinctly different area (north america instead of middle east, OECD-Europe instead of Russia)
	1.00	1.01	1.02		1.10
U5 Further technological	Data from enterprises, processes and materials under study (i.e. identical technology)		Data on related processes or materials but same technology, OR Data from processes and materials under study but from different technology	Data on related processes or materials but different technology, OR data on laboratory scale processes and same technology	Data on related processes or materials but on laboratory scale of different technology
	1.00		1.20	1.50	2.00
U6 Sample size	>100, continuous measurement, balance of purchased products	>20	> 10, aggregated figure in env. report	>=3	unknown
	1.00	1.02	1.05	1.10	1.20

Data Quality

$$SD_{g95} = \sigma_g^2 \\ = \exp \sqrt{(\ln(U_1))^2 + (\ln(U_2))^2 + (\ln(U_3))^2 + (\ln(U_4))^2 + (\ln(U_5))^2 + (\ln(U_6))^2 + (\ln(U_b))^2}$$

The factors U_1 till U_6 referring to the scores for:

- Reliability (U_1)
- Completeness (U_2)
- Temporal Correlation (U_3)
- Geographical Correlation (U_4)
- Further Technological (U_5)
- Sample Size (U_6)

The factor U_b refers to the basic uncertainty factor and is emission-specific.

Data quality

Data or assumption	Conventional	Electric	Qualitative description (rationale)
New Zealand electricity			For all descriptions relating to electricity below, refer also to Appendix E1.
Coal	(2,1,1,1,1,na)	(2,1,1,1,1,na)	Data from literature-source specific to NZ electricity, described in Section 3.5 and 4.3.
Water	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Estimation, based on AusLCI data
Black coal	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Assumes density of coal in NZ, based on AusLCI data
Transport (trucking)	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Estimation, based on AusLCI data
Transport (rail)	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Estimation, based on AusLCI data
Emissions to air (carbon dioxide, methane, dinitrogen monoxide, nitrogen oxides, carbon monoxide, NMVOC, sulfur dioxide)	(2,1,1,1,1,na)	(2,1,1,1,1,na)	Calculated, data from literature-source, based on total emissions from electricity generation using coal (2013 MfE data)
Emissions to air (other Australian NPI pollutants)	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Estimated, based on Australian NPI data for electricity from coal
Gas	(2,1,1,1,1,na)	(2,1,1,1,1,na)	Data from literature-source specific to NZ electricity, described in Section 3.5 and 4.3.
Water	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Estimation, based on AusLCI data
Natural gas	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Assumes carbon content of natural gas in NZ, based on AusLCI data
Transport (trucking)	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Estimation, based on AusLCI data
Transport (rail)	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Estimation, based on AusLCI data
Emissions to air (carbon dioxide, methane, dinitrogen monoxide, nitrogen oxides, carbon monoxide, NMVOC, sulfur dioxide)	(2,1,1,1,1,na)	(2,1,1,1,1,na)	Calculated, data from literature-source, based on total emissions from electricity generation using coal (2013 MfE data)
Emissions to air (Remaining Australian NPI pollutant data)	(3,3,1,3,1,na)	(3,3,1,3,1,na)	Estimated, based on Australian NPI data for electricity from coal
Hydropower	(2,1,1,1,1,na)	(2,1,1,1,1,na)	Literature source
Geothermal	(2,1,1,1,1,na)	(2,1,1,1,1,na)	Literature source